Exercise 3.5

1. Factorise the following expressions:
   (i) $2a^2 + 4a^2b + 8a^2c$ (ii) $ab - ac - mb + mc$

2. Factorise the following:
   (i) $x^2 + 4x + 4$ (ii) $3a^2 - 24ab + 48b^2$ (iii) $x^5 - 16x$
   (iv) $m^2 + \frac{1}{m^2} - 23$ (v) $6 - 216x^2$ (vi) $a^2 + \frac{1}{a^2} - 18$

3. Factorise the following:
   (i) $4x^2 + 9y^2 + 25z^2 + 12xy + 30yz + 20xz$
   (ii) $25x^2 + 4y^2 + 9z^2 - 20xy + 12yz - 30xz$

4. Factorise the following:
   (i) $8x^3 + 125y^3$ (ii) $27x^3 - 8y^3$ (iii) $a^6 - 64$

5. Factorise the following:
   (i) $x^3 + 8y^3 + 6xy - 1$ (ii) $l^3 - 8m^3 - 27n^3 - 18lmn$

3.5.2 Factorising the Quadratic Polynomial (Trinomial) of the type $ax^2 + bx + c$, $a \neq 0$

The linear factors of $ax^2 + bx + c$ will be in the form $(kx + m)$ and $(lx + n)$

Thus, $ax^2 + bx + c = (kx + m)(lx + n) = klx^2 + (lm + kn)x + mn$

Comparing coefficients of x^2, x and constant term c on both sides.
We have, $a = kl$, $b = (lm + kn)$ and $c = mn$, where ac is the product of kl and mn that is, equal to the product of lm and kn which are the coefficient of x. Therefore $(kl \times mn) = (lm \times kn)$.

Steps to be followed to factorise $ax^2 + bx + c$:

Step 1 : Multiply the coefficient of x^2 and constant term, that is ac.
Step 2 : Split ac into two numbers whose sum and product is equal to b and ac respectively.
Step 3 : The terms are grouped into two pairs and factorise.

Example 3.27

Factorise $2x^2 + 15x + 27$

Solution

Compare with $ax^2 + bx + c$
we get, $a = 2$, $b = 15$, $c = 27$
product $ac = 2 \times 27 = 54$ and sum $b = 15$
We find the pair 6, 9 only satisfies "$b = 15$"

Product of numbers $ac = 54$	Sum of numbers $b = 15$	Product of numbers $ac = 54$	Sum of numbers $b = 15$
1×54	55	-1×-54	-55
2×27	29	-2×-27	-29
3×18	21	-3×-18	-21
6×9	**15**	-6×-9	-15
The required factors are **6** and **9**			

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